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BI-5050 — Midterm Study Guide

Exam Format: 2 hours | Practical Excel Exam | No open book or notes

The midterm will focus on your ability to build and use spreadsheet models based on the techniques demonstrated in class and practiced in your assignments, and the ability to interpret results.

Mathematical formulas will be provided as needed. You will be expected to translate them into working Excel models.

Topics to Review:

1. Spreadsheet Modeling

- Organize data, calculations, outputs, and a summary sheet for major results.
- Build models for auditability using named ranges as appropriate.
- Create models that respond correctly to changes in inputs.
- Provide examples of good spreadsheet model engineering practices.

2. Break-Even and Discounted Cash Flow Models

- Build a production break-even model.
- Build a multiyear DCF model with growth rates, forecasts, random errors, and NPV calculations.

3. What-If and Sensitivity Analysis

- Create switches to turn forecast errors on/off or select Base Case versus What-If inputs.
- Perform one-variable sensitivity analysis and construct tornado charts.

4. Data Cleaning with Power Query

- Import and clean messy data using Power Query.
- Correct data types, missing values, duplicates, and inconsistent formatting.

5. Statistics and Charts

- Calculate and interpret descriptive statistics, correlation, and regression.
- Create appropriate summary charts, including bar charts and scatter plots.
- Present and interpret results for management.

Preparing for the Exam

Review my class demos (videos, class recordings, demo sheets) and your completed Excel assignments. Practice **building and modifying models**. This is about applying what you've learned.

Interpret your results and explain their business implications so a non-technical manager can understand their significance. Why should a manager care about your numbers?

- Bad — Technical:
  - "A ±10% change in annual transactions produces an NPV range of –$45,000 to $210,000."
- Good — Management:
  - "Store profitability is sensitive to customer traffic. A 10% decrease in transactions could make this store unprofitable, so our sales forecast is a major driver of the decision."